CIP-B101 Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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C11/26/PCDFIR/17007

Course code:

CIP-B101, Basic Computer Skills for Digital Forensics

Lab title:

Lab 1B, PC System Fundamentals, Disk Sectors and Partition Analysis

Date:

6 November 2026

Instructor's name

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1. Introduction

The aim of this lab was to check and understand the basic details of a computer that could be used in a digital forensic investigation, without making any changes to it.

I started by checking the computer's hardware and operating system. I also looked at the different places where digital evidence can be found, checked the processes running on the computer and inspected the disk and its partitions using commands that only display information.

After that, I worked on calculating partition sizes from their sector counts, explaining how a computer boots, comparing MBR and GPT, and identifying the file systems on the disk.

Throughout the lab, I only used commands to view information. I did not format, delete, resize or create anything on the computer.

2. Tools Used

Tool	**What I used it for**
Windows 11 Pro (my HP EliteBook 840 G6)	Checking system information, storage, partitions and file systems
Msinfo32 and systeminfo	Checking system specifications, BIOS mode and Secure Boot
PowerShell (Get-Disk, Get-Partition. Get-Volume, Get-Process)	Viewing disk, partition, volume and process information
Command Prompt (tasklist, DiskPart)	Viewing running processes and listing disks, partitions and volumes
Disk Management (diskmgmt.msc)	Viewing the disk layout
PowerShell as a calculator	Checking my sector and partition-size calculations

I also set my command prompts to show my name (utibe_eshiet), so that my screenshots can be linked to my work.

3. System Specification Evidence

I started by using the systeminfo command to get information about my computer and the output I got was.

None

Host Name: DESKTOP-5SGSMVO

OS Name: Microsoft Windows 11 Pro

OS Version: 10.0.26200 N/A Build 26200

System Manufacturer: HP

System Model: HP EliteBook 840 G6

System Type: x64-based PC

Processor(s): Intel64 Family 6 Model 142 Stepping 12 GenuineIntel ~1596 Mhz

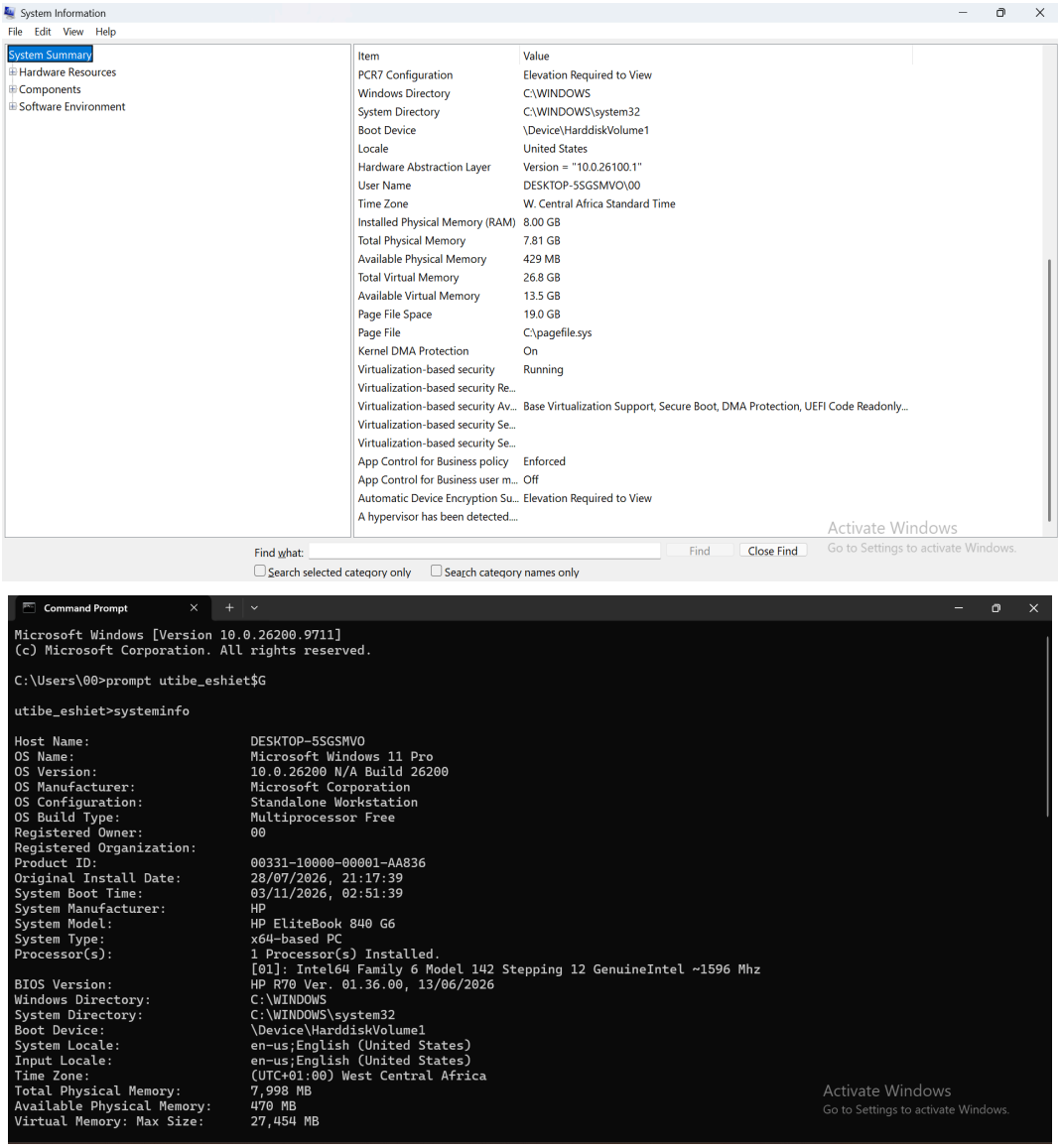
BIOS Version: HP R70 Ver. 01.36.00, 13/06/2026

Boot Device: \Device\HarddiskVolume1

Time Zone: (UTC+01:00) West Central Africa

Total Physical Memory: 7,998 MB

Page File Location(s): C:\pagefile.sys



SCREENSHOT 1: msinfo32 System Summary and the systeminfo

I also checked the system information using msinfo32. These are the details I recorded:

Item	**Value from my System**
OS Name and Version	Microsoft Windows 11 Pro, version 10.0.26200
System Type	x64-based PC (64-bit)
Processor	Intel Core i5-8365U @ 1.60GHz, 4 cores, 8 logical processors
BIOS Version	HP R70 Ver. 01.36.00(13/06/2026)
BIOS Mode	UEFI

Secure Boot State	On
Installed Physical Memory	8.00 GB

From the results, I confirmed that my computer is a 64-bit system and uses UEFI mode, with Secure Boot turned on. The boot device is listed as HarddiskVolume1, which is the first partition on the disk. I also noticed the page file location, C:\pagefile.sys. This is where Windows stores some information when it needs extra memory, so it may contain traces that could be useful during a forensic investigation.

Another thing I noted was the time zone, which is UTC+01:00. This is important when working with timestamps because I would need to consider the time difference when converting local times to UTC. I also noticed a VirtualBox Host-Only network adapter in the network information. This shows that virtualization software is installed on the computer.

4. Storage Information

I first tried to use wmic, but it was not available on my version of Windows 11. So, I used the PowerShell alternative allowed by the lab.

Command used:

```
None
utibe_eshiet> Get-Disk
```

Output:

```
None
Number Friendly Name  Serial Number  HealthStatus  OperationalStatus  Total Size  Partition Style

0      SAMSUNG MZ...  (hidden)       Healthy       Online             238.47 GB   GPT
```

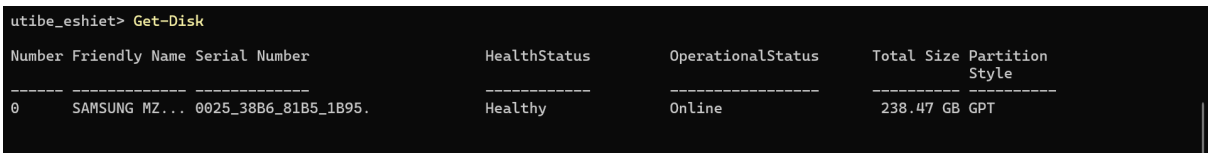
I also used Get-Volume to check the volumes on the computer.

Output:

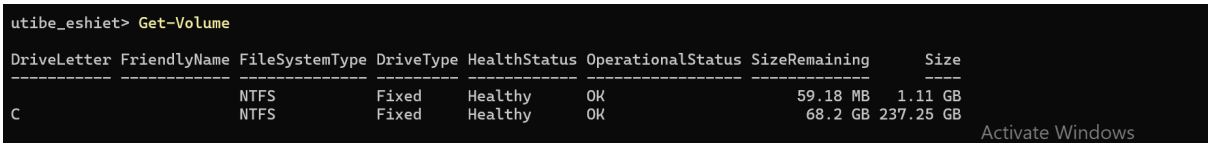
```
None
DriveLetter FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size

            NTFS           Fixed     Healthy      OK                    59.18 MB    1.11 GB

C           NTFS           Fixed     Healthy      OK                    68.2 GB  237.25 GB
```



SCREENSHOT 2: Get-Disk output showing the disk model and size



SCREENSHOT 3: Get-Volume output showing the logical volumes and file systems

From the results, I found that my computer has one Samsung NVMe SSD, model MZFLW256HEHP. Windows reports its size as 238.47 GB, and the disk uses the GPT partition style. The volume information also showed the available volumes and their file system details.

5. Evidence Layer Mapping

Digital evidence can be found in different parts of a computer. For this lab, I looked at five layers and the kind of evidence that may be found in each one.

Layer/Source	**Example Evidence**	**Where it may be found**	**Why it matters**
User layer	Opened files, searched terms, typed usernames	Recent files, browser history, Powershell history	Can link a person's actions to activity on the computer
Application layer	Browser cache, Office document metadata, chat logs	Application folders and databases in the user profile	Can show which software created or used the data
OS layer	Even logs, installed programs, user accounts	Windows Event Logs, the registry, C:\Windows	Can show system events and account activity
Disk layer	Deleted files, file fragments, partitions	NTFS on C:, unallocated space and partitions on the Samsung disk	Can help with file recovery and timeline analysis
Memory/process layer	Running processes, PIDs, network connections	RAM (8 GB) and C:\pagefile.sys	Can show activity that may disappear after shutdown

Why evidence may exist in more than one layer

One action by a user can leave traces in different parts of a computer.

For example, when someone opens a document, it may appear in the Recent Files list. The application used to open it may also create temporary or autosave files. Windows may record related activity in the registry or event logs, while the actual document and its timestamps are stored on the disk. The contents of the document may also remain in the computer's memory while it is open.

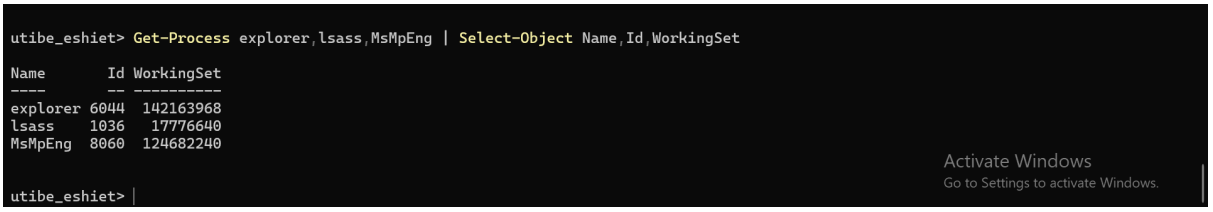
This means that an examiner does not have to depend on just one source of evidence. They can compare information from different layers to get a clearer picture of what happened. Even if one trace is deleted, another one may still be available.

6. Process Observation

For this part, I used the tasklist command to see the processes running on my computer.

Output:

```
None
Image Name                     PID Session Name        Session#    Mem Usage
explorer.exe                  6044 Console                    1    183,276 K
lsass.exe                     1036 Services                   0     20,604 K
MsMpEng.exe                   8060 Services                   0    208,252 K
(shortened: the full list has more than 250 processes)
```



SCREENSHOT 4: tasklist and Get-Process output showing PIDs

I picked three processes from the list and looked at what they do.

Process	**PID**	**Likely Purpose**
explorer.exe	6044	This is the Windows shell. It manages the desktop, taskbar and File Explorer. It runs in the logged-in user's section and is involved when the user opens programs.
lsass.exe	1036	This is the Local Security Authority process. It handles logins and checks user

		credentials. Attackers may target it to steal passwords, so examiners may check it for signs of tampering.
MsMpEng.exe	8060	This is the Microsoft Defender antimalware engine. Its presence shows that Defender was running, and its logs may contain information about detected threats.

The PIDs shown above belong to this particular capture. They can change when the computer restarts, so the numbers may be different the next time I check.

Difference between a program, process and thread

A program is a file stored on the computer, while a process is that program running. A thread is a smaller unit of work within a process. It shares the process's memory and helps the program carry out different tasks.

Checking processes is useful in digital forensics, especially when investigating malware. It helps an examiner see what was running on the computer at the time of the investigation. This is important because some malware may only exist in memory and could disappear when the computer is shut down.

7. Disk Sector and Disk Size Calculation

For this part, I worked with the formula for calculating disk or partition size.

The formula is: Disk or partition size = Number of sectors × Bytes per sector

To convert the answer from bytes to GB, I divided by 1024 three times.

7.1 Slide calculation

The values given were:

- Sectors per track = 400
- Heads = 12
- Cylinders = 17,000
- Bytes per sector = 512

I used these values to calculate the total number of sectors and the disk size.

Item	Working	Answer
Total sectors	400 x 12 x 17,000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000
Kilobytes	41,779,200,00 ÷ 1024	40,800,000 KB
Megabytes	40,800,000 ÷ 1024	39,843.75 MB
Gigabytes	39.843.75 ÷ 1024	38.91 GB

I also checked the calculation in PowerShell. It returned 38.909912109375, which rounds up to 38.91 GB.

8. Disk and Partition Inspection

Next, I checked the disk and its partitions using PowerShell and below is the output:

```
None
Number             : 0

FriendlyName       : SAMSUNG MZFLW256HEHP-000MV

PartitionStyle     : GPT

LogicalSectorSize  : 512

PhysicalSectorSize : 4096

Size               : 256060514304
```

I then used another command to list the partitions on the disk.

Command used:

```
None
utibe_eshiet> Get-Partition -DiskNumber 0 | Select-Object PartitionNumber,Type,Offset,Size
```

Output:

```
None
PartitionNumber Type          Offset        Size

              1 System        1048576    104857600

              2 Reserved    105906176     16777216
```

```
3 Basic       122683392 254747344896
4 Recovery 254870028288   1188036608
```

I also checked the disk using DiskPart.

Commands and output:

```
None
DISKPART> list disk

  Disk ###  Status        Size     Free     Dyn  Gpt
  Disk 0    Online        238 GB  2048 KB        *

DISKPART> list partition

  Partition ###  Type            Size     Offset
  Partition 1    System           100 MB  1024 KB
  Partition 2    Reserved          16 MB   101 MB
  Partition 3    Primary          237 GB   117 MB
  Partition 4    Recovery        1133 MB   237 GB
```



SCREENSHOT 5: Disk Management, Get-Partition and DiskPart output

From the results, I confirmed that the disk uses GPT. DiskPart shows a star under the Gpt column for Disk 0, and PowerShell also identifies the partition style as GPT.

I also found that the logical sector size is 512 bytes, while the physical sector size is 4096 bytes. For my partition calculations, I used the logical sector size of 512 bytes. Both PowerShell and DiskPart showed four partitions on the disk.

9. Manual Partition Calculations

For this part, I calculated the start sector, sector count, end sector and size of the partitions.

These are the formulas I used:

```
None
Start sector = Offset ÷ 512


Sector count = Size in bytes ÷ 512


End sector = Start sector + Sector count - 1


Size in GB = Sectors × 512 ÷ 1024 ÷ 1024 ÷ 1024
```

Partition 1 and Partition 2 calculations

Calculation	Partition 1 (System)	Partition 2 (Reserved)
Start Sector	1,048,576 ÷ 512 = 2,048	105,906,176 ÷ 512 = 206,848
Sector count	104,857,600 ÷ 512 = 204,800	16,777,216 ÷ 512 = 32,768
End sector	2,048 + 204,3800 - 1 =206,847	206,848 + 32,768 - 1 = 239,615
Size in bytes	204,800 × 512 = 104,857,600	32,768 × 512 = 16,777,216
Size in GB	104,857,600 ÷ 1024 ÷ 1024 ÷ 1024 = 0.0977 GB	16,777,216 ÷ 1024 ÷ 1024 ÷ 1024 = 0.0156 GB

```
utibe_eshiet> Get-Partition -DiskNumber 0 | Select-Object PartitionNumber, Type, Offset, Size, @{n='StartSector';e={$_.Offset/512}},
@{n='Sectors';e={$_.Size/512}}, @{n='EndSector';e={$_.Offset/512 + $_.Size/512 - 1}}, @{n='GB';e={[math]::Round($_.Size/1GB,4)}}


PartitionNumber : 1
Type            : System
Offset          : 1048576
Size            : 104857600
StartSector     : 2048
Sectors         : 204800
EndSector       : 206847
GB              : 0.0977
```

SCREENSHOT 6: partition 1 calculation evidence (sector calculation output)

```
PartitionNumber : 2
Type            : Reserved
Offset          : 105906176
Size            : 16777216
StartSector     : 206848
Sectors         : 32768
EndSector       : 239615
GB              : 0.0156
```

SCREENSHOT 7: partition 2 calculation evidence (sector calculation output)

I also checked all four partitions using PowerShell. The command divided each partition's offset and size by 512 to give me the sector information.

No.	Type	Start Sector	Sectors	End sector	Size (GB)
1	System	2,048	204,800	206,847	0.0977
2	Reserved	206,848	32,768	239,615	0.0156
3	Basic (C:)	239,616	497,553,408	497,793,023	237.252
4	Recovery	497,793,024	2,320,384	500,113,407	1.1064

From the calculations, I noticed that each partition starts one sector after the previous partition ends. This means there are no gaps or overlaps between the partitions.

Partition 1 starts at sector 2,048 instead of sector 0 because the beginning of a GPT disk contains information such as the protective MBR, GPT header and partition entries. Windows also aligns the first partition at the 1 MB boundary.

Part G: Final Practical Assignment

Task	Required evidence	My answer
Identity partition 1	Partition table and command output	Partition 1 is the System partition (EFI system partition). It uses FAT32 and has an offset of 1,048,576 bytes
Partition 1 sector count	Start, end or sector count	Start: 2,048; end: 206,847; total:204,800 sectors.
Partition 1 size	Formula and result in GB	204,800 × 512 ÷ 1024 ÷ 1024 ÷ 1024 = 0.0977 GB

		(100 MB)
Identify partition 2	Partition table and command output	Partition 2 is the Reserved partition (Microsoft Reserved Partition), with an offset of 105,906,176 bytes
Partition 2 sector count	Start, end or sector count	Start: 206,848; end: 239,615; total: 32,768 sectors.
Partition 2 size	Formula and result in GB	32,768 × 512 ÷ 1024 ÷ 1024 ÷ 1024 = 0.0156 GB (16 MB)
Compare with OS-reported size	Disk Management, Get-Partition or DiskPart output	Windows reports 100 MB for partition 1 and 16 MB for partition 2
Explain difference if any	Rounding, reserved space, overhead, GiB/GB	There is no significant difference. Windows displays the sizes rounded to whole MB, while my calculations use binary units.

Comparing my calculations with the sizes reported by Windows

For partition 3, my calculation gave me 237.25 GB, which is the same as the size reported by Get-Volume. For partition 4, my calculation gave me 1.1064 GB, while Get-Volume reported 1.11 GB. This is a small difference caused by how the values are displayed and rounded.

I also noticed that the partition sizes in bytes (254,747,344,896 and 1,188,036,608) are 4,096 bytes larger than the volume sizes reported by Get-Volume (254,747,340,800 and 1,188,032,512). DiskPart displays the sizes as whole numbers, which is why the main partition appears as 237 GB.

Another thing I noticed is that the SSD is sold as a 256 GB drive, but Windows displays it as 238.47 GB. This is because manufacturers use decimal units when stating disk capacity, while Windows uses binary units and divides by 1024 during the conversion.

10. Boot Process Explanation

My laptop uses UEFI mode. I confirmed this by checking msinfo32, which showed BIOS Mode as UEFI and Secure Boot State as On. The boot device shown in systeminfo is HarddiskVolume1. This corresponds to partition 1, which is the 100 MB System partition.

These are the main stages that happen when a computer starts:

1. Power-on: When I switch on the laptop, the power supply starts and the processor begins running the firmware code.
2. POST (Power-On Self-Test): The computer checks important hardware components such as the processor, RAM and connected devices to make sure they are working.
3. UEFI firmware and CMOS settings: The firmware reads the saved settings, including the boot order and Secure Boot setting. It then decides which device to boot from.
4. GPT and EFI boot record: The firmware checks the GPT partition information and locates the EFI System Partition. On my laptop, this is partition 1, where the boot files are stored.
5. Boot manager and loader: Windows Boot Manager starts, and the Windows loader prepares the operating system.
6. Kernel loading: The Windows kernel and the required boot drivers are loaded into memory.
7. Operating system control: Windows finishes starting, its services begin running and the user can log in. System logs and other user activity also begin to build up.

The boot process and its forensic relevance

Stage	**What happens**	**Forensic relevance**
Power-on and POST	Hardware is started and checked	Hardware failure or tampering may affect access to evidence
BIOS/UEFI firmware	Boot order and firmware settings are applied.	The boot mode helps explain the partition style and boot information
MBR/GPT or EFI boot	The computer finds the bootable partition or EFI boot entry	Boot records may provide information about the boot configuration
Boot loader/manager	The operating system is selected and prepared to start	This may help identify multiple operating systems or hidden OS behaviour.
Kernel and OS control	The operating system finishes starting	System logs and user activity begin to accumulate



OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-5SGSMVO
System Manufacturer	HP
System Model	HP EliteBook 840 G6
System Type	x64-based PC
System SKU	4WG30AV
Processor	Intel(R) Core(TM) i5-8365U CPU @ 1.60GHz, 1896 Mhz, 4 Core(s), 8 Logical Pro...
BIOS Version/Date	HP R70 Ver. 01.36.00, 13/06/2026
SMBIOS Version	3.1
Embedded Controller Version	82.113
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	8549
BaseBoard Version	KBC Version 52.71.00
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	DESKTOP-5SGSMVO\00
Time Zone	W. Central Africa Standard Time

SCREENSHOT 8: msinfo32 showing BIOS Mode (UEFI) and Secure Boot State

MBR and GPT

MBR and GPT are two different ways of organising partitions on a disk.

Feature	MBR	GPT (used on my disk)
Age / design	An older partitioning system from the 1980s, commonly used with legacy BIOS.	A more modern partitioning system commonly used with UEFI.
Size limitation	Supports disks of about 2 TB.	Supports very large disks. The slide notes about 9.4 ZB.
Partition information	Stored in the first sector together with the boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Important information is stored in sector 0, so damage to that area can affect how the partitions are read.	The backup structures can help with recovery and checking the partition information.

The main difference I noticed is that GPT is the partition style used on my laptop. It also keeps backup partition information, which can be useful if the main GPT information is damaged.

11. File System Identification

For this part, I used PowerShell and DiskPart to check the file systems on the volumes.

Output:

```
None
DriveLetter FileSystemType        Size

            NTFS            1188032512

C           NTFS            254747340800
```

I also used DiskPart to list the volumes.

Command and output:

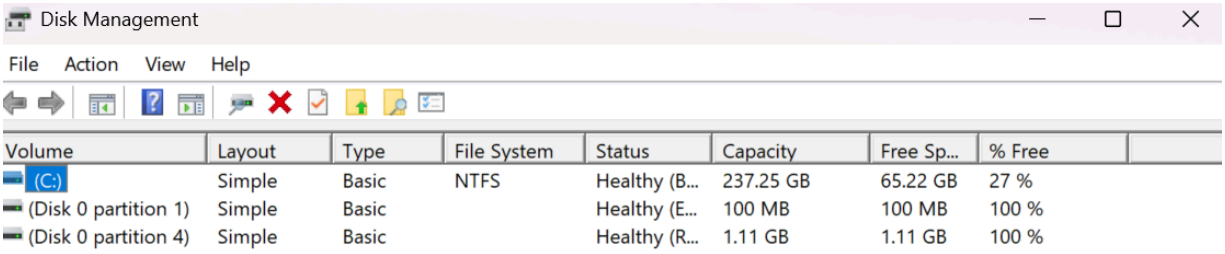
```
None
DISKPART> list volume

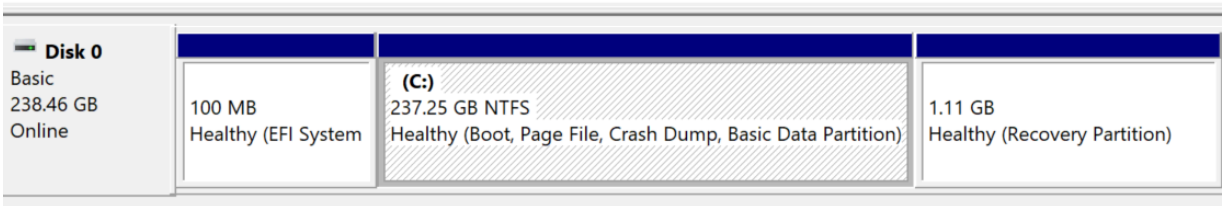

  Volume ###  Ltr  Label  Fs     Type        Size     Status    Info

  Volume 0     C         NTFS   Partition    237 GB  Healthy  Boot

  Volume 1              FAT32  Partition    100 MB  Healthy  System

  Volume 2              NTFS   Partition   1133 MB  Healthy  Hidden
```





SCREENSHOT 9: Get-Volume and DiskPart list volume showing the file systems

From the DiskPart results, I identified the following volumes:

Volume	Partition	Size	File system	Role
Volume 0 (C:)	Partition 3	237 GB	NTFS	Contains Windows and user data. It is the boot volume
Volume 1	Partition 1	100 MB	FAT32	The EFI System, Partition. It contains the boot files
Volume 2	Partition 4	1133 MB	NTFS	Hidden recovery parttion

Partition 2, which is the 16 MB Reserved partition, does not have a file system volume of its own.

Why a file system is important in digital forensics

A file system controls how information is stored and organised on a partition. It determines how files are named, where they are stored, how they are located and what information is kept about them.

For example, my C: drive uses NTFS. It stores information such as MFT entries, timestamps, permissions and a journal. These details can help an examiner understand when files were created or changed, build timelines and sometimes recover deleted files.

FAT32 is a simpler file system that is supported by many devices. This is why the computer's firmware can read it during the boot process. It is also commonly used on USB drives.

Knowing the file system is important because it helps an examiner decide which tools to use and what kind of evidence or file information they should look for.

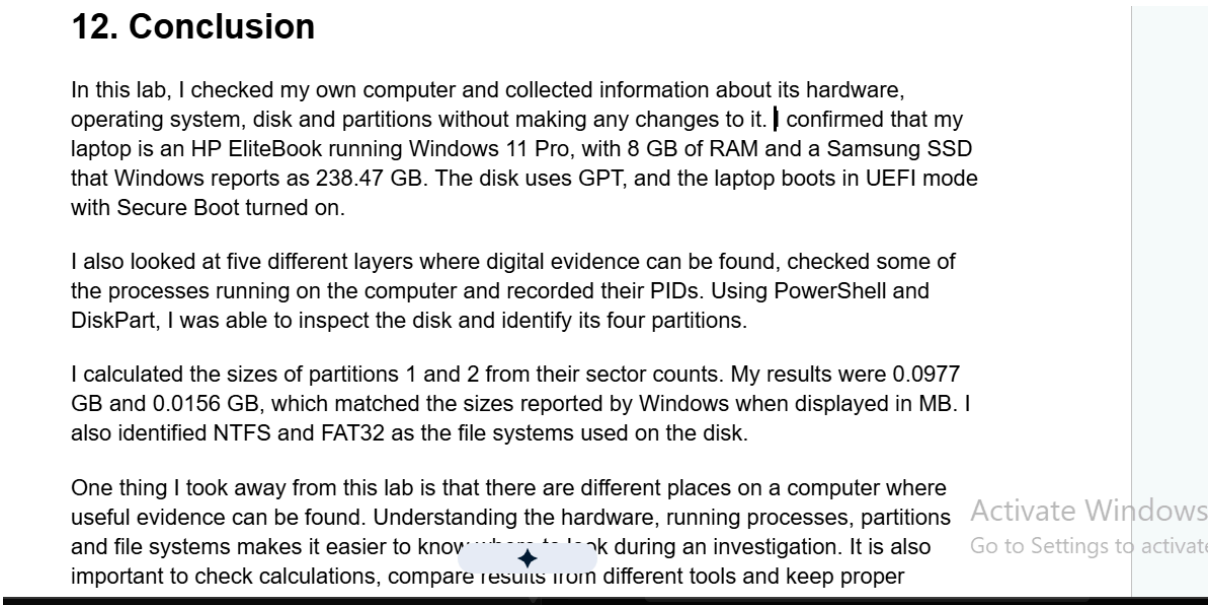
12. Conclusion

In this lab, I checked my own computer and collected information about its hardware, operating system, disk and partitions without making any changes to it. I confirmed that my laptop is an HP EliteBook running Windows 11 Pro, with 8 GB of RAM and a Samsung SSD that Windows reports as 238.47 GB. The disk uses GPT, and the laptop boots in UEFI mode with Secure Boot turned on.

I also looked at five different layers where digital evidence can be found, checked some of the processes running on the computer and recorded their PIDs. Using PowerShell and DiskPart, I was able to inspect the disk and identify its four partitions.

I calculated the sizes of partitions 1 and 2 from their sector counts. My results were 0.0977 GB and 0.0156 GB, which matched the sizes reported by Windows when displayed in MB. I also identified NTFS and FAT32 as the file systems used on the disk.

One thing I took away from this lab is that there are different places on a computer where useful evidence can be found. Understanding the hardware, running processes, partitions and file systems makes it easier to know where to look during an investigation. It is also important to check calculations, compare results from different tools and keep proper records of the work without changing the original system.



12. Conclusion

In this lab, I checked my own computer and collected information about its hardware, operating system, disk and partitions without making any changes to it. I confirmed that my laptop is an HP EliteBook running Windows 11 Pro, with 8 GB of RAM and a Samsung SSD that Windows reports as 238.47 GB. The disk uses GPT, and the laptop boots in UEFI mode with Secure Boot turned on.

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Activate Windows
Go to Settings to activate

SCREENSHOT 10: a screenshot of the conclusion page

13. Screenshot Checklist

Before submitting, I checked that I had included the screenshots required for the lab.

No	Screenshot required	Included?
1	System Information or Systeminfo output	Yes
2	Disk drive model and sizge output	Yes
3	Logical volume/file system output	Yes
4	Running process/PID output	Yes
5	Disk Management, Get-Parttion, DiskPart, lsblk, fdisk or parted output	Yes
6	Partition 1 calculation evidence	Yes
7	Partition 2 calculation evidence	Yes
8	Boot mode evidence: BIOS/UEFI	Yes
9	File system identification evidence	Yes

10	Final report summary/conclusion page	Yes